SQL Fundamental

Exercise 3 : SQL CASE STATEMENTS

1) Classifying

```
SELECT    product_id,
          product_name,
          price
  CASE
        WHEN  price > 1000 THEN 'Expensive'
        WHEN  price BETWEEN 100 AND 1000 THEN 'Mid-range'
        ELSE  'Budget'
  END AS price_category
FROM  products
```

product

product_id	product_name	price	price_category
1	Laptop	1200	Expensive
2	Phone	800	Mid_range
3	Keyboard	45	Budget
4	Monitor	300	Mid_range
5	Mouse	25	Budget

2)

```
SELECT    customer_name,
          amount,
  CASE
        WHEN amount ≥ 1000 THEN 'High_value'
        WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium_value'
        ELSE 'Low value'
  END AS order_value_category
FROM  orders
```

orders

customer_name	amount	order_value_category
Alice	150.00	Low_value
Bob	560.00	Medium_value
Charlie	999.99	Medium_value
Dana	45.50	Low_value
Ethan	1200.00	High_value

3)
```
SELECT emp_name,
       department,
       salary,
  CASE
    WHEN department = 'IT' AND salary > 80000 THEN Senior IT
    WHEN department = 'HR' AND salary > 55000 THEN Experienced HR
    ELSE 'staff'
  END AS position_level
FROM employees;
```

employees

emp_name	department	salary	position_level
John	IT	85 000	Senior_IT
Sara	HR	60 000	ExperiencedHR
Mark	IT	75000	staff
Lucy	Finance	95000	staff
Tom	HR	55000	Experienced HR

4)
```
SELECT student_name,
       score,
  CASE BY
    WHEN SCORE ≥ 90 THEN 'A'
    WHEN SCORE BETWEEN 80 AND 89 THEN B
    WHEN SCORE BETWEEN 70 AND 79 THEN C
    WHEN SCORE BETWEEN 60 AND 69 THEN D
    ELSE F
    AND AS grade
  FROM students
```

students

student name	score	Grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5)
```
SELECT *
    CASE BY
        WHEN delivery_time_minute <30 THEN 'FAST'
        WHEN delivery_time BETWEEN 30 AND 60 THEN 'ON TIME'
        ELSE 'LATE'
    END AS PERFOMANCE
FROM deliveries
```

deliveries

Delivery_ID	delivery time minute	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6
```
SELECT issue_type,
       property,
    CASE BY
        WHEN priority = 1 THEN 'LOW'
        WHEN priority = 2 THEN 'MEDIUM'
        WHEN priority = 3 THEN 'HIGH'
    END AS priority_label
FROM tickets
```

Issue_type	priority	priority_label
Login_issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7)

```
SELECT student_id,
       attendance_percentage,
       CASE BY
          WHEN (days_pressent ÷ total_days × 100) ≥ 90% THEN 'EXCELLENT'
          WHEN (dys_present ÷ total_days × 100) BETWEEN 75 AND 89 THEN 'GOOD'
          ELSE 'NEEDS IMPROVEMENT'
       END AS attendance_status
FROM attendance status;
```

attendance

student_id	days_present	~~total_days~~ attendance status
1	45	EXCELLENT
2	30	NEEDS IMPROVEMENT
3	48	EXCELLENT
4	25	NEEDS IMPROVEMENT
5	50	EXCELLENT

8

```
SELECT *
CASE BY
   WHEN stock_qty = 0 THEN 'Out of stock'
   WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
   ELSE 'In Stock'
END AS stock_status
FROM products_inventory.
```

products - inventory

product_id	stock_qty	stock_status
1	5	low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

```
SELECT subject,
       enrolled students,
  CASE BY
    WHEN enrolled_students ≥ 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    WHEN enrolled_students < 10 THEN 'Small'
  END AS class_size_category
FROM classes
```

classes

subject	enrolled_student	class_size category
Maths	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10) SELECT *

CASE BY

WHEN payment method = Cash AND amount ≥ 200

THEN ELIGIBLE FOR DISCOUNT

ELSE Not Eligible

FROM payments.

payments

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount